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COMPUTER SCIENCE AND ENGINEERING

Design Document

Students&Companies

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Contents

1	Introduction	4
1.1	Purpose	4
1.2	Scope	5
1.3	Definitions, Acronyms, Abbreviations	6
1.4	Revision History	6
1.5	Reference Documents	7
1.6	Document Structure	7
2	Architectural Design	9
2.1	Overview: High-level components and their interaction	9
2.2	Component View	10
2.2.1	Students&Companies Server (S&C Server)	10
2.2.2	Model	10
2.2.3	Notification Manager	11
2.2.4	Account Manager	11
2.2.5	AuthManager	11
2.2.6	Application Manager	11
2.2.7	Offer Manager	12
2.2.8	Recommendation Manager	12
2.2.9	Selection Process Manager	14
2.2.10	Internship Monitoring & Complaint Manager	15
2.2.11	Feedback & Statistics Manager	15
2.2.12	CV & Offer Advisor	16
2.3	Deployment View	16
2.4	Runtime View	18
2.4.1	THE STUDENT REGISTERS ON THE PLATFORM	18
2.4.2	STUDENT'S CV CREATION AND IMPROVEMENT	19
2.4.3	COMPANY REGISTRATION	20
2.4.4	POSTING INTERNSHIP POSITIONS	21
2.4.5	RECOMMENDATION AND MATCHING SYSTEM (MATCHING COMPANIES AND STUDENTS)	22
2.4.6	STUDENT'S SPONTANEOUS APPLICATION	23
2.4.7	INTERNSHIP WITHDRAWAL AND APPLICATION CANCELLATION PROCESS	24
2.4.8	INTERNSHIP APPLICATION REVIEW AND SE- LECTION PROCESS	25
2.4.9	THE STUDENT HAS BEEN SELECTED FOR AN INTERNSHIP IN A COMPANY	26

2.4.10	MONITORING OF INTERNSHIPS BY THE UNIVERSITY	27
2.4.11	MANAGEMENT OF REPORTS AND COMPLAINTS BY THE UNIVERSITY	28
2.4.12	CV IMPROVEMENTS FOR STUDENTS	30
2.4.13	SUGGESTIONS FOR COMPANY	31
2.4.14	FEEDBACK AND STATISTICS COLLECTION	32
2.4.15	MODIFY STUDENT PROFILE TO CHANGE INTERNSHIP PREFERENCES	33
2.5	Component Interfaces	34
2.6	Selected Architectural Styles and Patterns	40
2.7	Other Design Decisions	41
2.7.1	Availability	41
2.7.2	Security	41
3	User Interface Design	42
4	Requirements Traceability	53
5	Implementation, Integration and Test Plan	58
5.1	General Implementation	58
5.2	Database Integration	58
5.2.1	Authentication	58
5.2.2	University Client	59
5.2.3	Student and Company Advanced Functionalities	60
5.3	Test Plan	61
5.3.1	General Testing	61
5.3.2	AI Training	62
6	Effort Spent	63

1 Introduction

1.1 Purpose

The purpose of this document is to present a comprehensive and detailed Design Description of the **Students&Companies** system, illustrating the key design decisions made during its development. This document provides a complete overview of the system, covering topics such as the general architectural layout, the design of specific components and their interfaces, the physical deployment of components, and a presentation of the user graphical

interfaces. Additionally, it includes a discussion of the Implementation and Testing plan.

The Design Document builds upon the requirements and use cases discussed in the Requirement Analysis and Specification Document (**RASD**), demonstrating how these are implemented in the system. It serves as a critical resource to ensure alignment between the design and implementation phases. This document is intended for software developers and testers, offering them a roadmap to guide the development, integration, and verification processes while ensuring consistency and quality throughout the system's lifecycle.

1.2 Scope

The **Students&Companies** platform is designed to facilitate the process of matching university students seeking internships with companies offering them. The platform provides tailored functionalities to support the specific needs of each user role: students, companies, and universities.

Students have access to a Student HomePage, where they can browse available internship opportunities, apply for those of interest, and track the status of their applications. Through the CV Management section, located in their Profile Section, students can upload and update their CVs to ensure their profiles remain attractive to recruiters. Additionally, the platform proactively notifies students of new internship opportunities that match their skills and preferences via the Recommendations feature, delivering updates through in-app notifications.

Companies utilize the platform via the Company HomePage, which enables them to post new internship offers and monitor applications received. The Internship Management section provides tools for reviewing student applications, managing ongoing internships, scheduling interviews, and collecting responses through structured questionnaires. Companies also benefit from feedback provided by the Recommendation feature, which offers suggestions for optimizing internship postings to attract better matches.

Both students and companies can access a Complaint Management section to alert universities about issues related to internships. This section serves as a centralized space for reporting and addressing concerns.

Universities interact with the platform through a University HomePage, where they can oversee all internships involving their students. The Complaint Handling functionality allows universities to address issues that may

arise during internships and, if necessary, communicate with both parties to facilitate resolutions or even terminate the internship. Additionally, universities can access reports and insights regarding the performance and status of ongoing internships.

The platform also features a Recommendation System to match students with internships based on their skills, experiences, and preferences. It employs a range of matching mechanisms, from keyword-based searches to advanced statistical analyses, to ensure optimal results. By gathering feedback from both students and companies, the system continuously refines its algorithms to enhance the overall user experience.

In summary, the **S&C platform** streamlines the entire internship lifecycle from matching students with companies to managing ongoing internships creating an efficient, user friendly system that meets the needs of all stakeholders.

1.3 Definitions, Acronyms, Abbreviations

This section contains a lists of all the relevant definitions, acronyms, and abbreviations used throughout the document.

Acronyms

- I. S&C: Students&Companies
- II. CV: Curriculum Vitae
- III. R_i : Requirement $i - th$
- IV. RASD: Requirement Analysis and Specification Document

1.4 Revision History

Version	Date	Changes
0.1	12/19/2024	Start&End Introduction
0.2	12/22/2024	Start Architectural Design + User Interface Design

Version	Date	Changes
0.3	12/23/2024	Work on Architectural Design + User Interface Design
0.4	12/24/2024	Start Requirements Traceability
0.5	12/27/2024	Start Implementation, Integration, and Test Plan
0.6	12/28/2024	End Requirements Traceability + End Implementation, Integration, and Test Plan.
0.7	12/30/2024	Rewiev.
0.8	12/31/2024	Rewiev.
0.9	01/2/2025	End User Interface Design.
1.0	01/4/2025	Final Recap.

1.5 Reference Documents

The specification document Assignment RDD AY 2024-2025.pdf

1.6 Document Structure

This document is structured as follows:

- Section 2: Architectural Design:**
 This section provides a comprehensive overview of the system and sub-system architecture. It explains how the various subsystems interact with each other to ensure all functionalities required by the project are delivered.
- Section 3: User Interface Design**
 This part describes the system's functionalities from the user's perspective, showcasing all graphical interfaces. The goal is to provide a clear understanding of how users will interact with the application.
- Section 4: Requirements Traceability**
 This section analyzes how the requirements defined in the RASD document are mapped to the design elements outlined in the architectural design section, ensuring consistency between the specifications and the implementation.

- **Section 5: Implementation, Integration, and Test Plan**

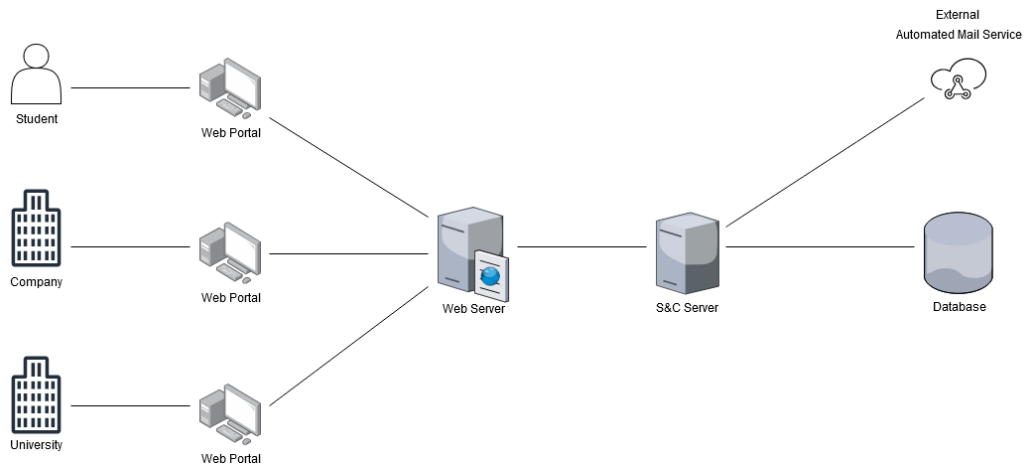
This section details the entire development process, from implementation to integration, and testing, ensuring that each component operates as expected.

- **Section 6: Effort Spent**

This part keeps track of the time dedicated to drafting and completing this document.

2 Architectural Design

2.1 Overview: High-level components and their interaction



The system can be divided into two main components:

1. Client Side:

- **Web Portal:** All users will access the platform's features through the S&C web portal. Students, company managers, and university representatives will log in using their credentials to utilize the available functionalities.

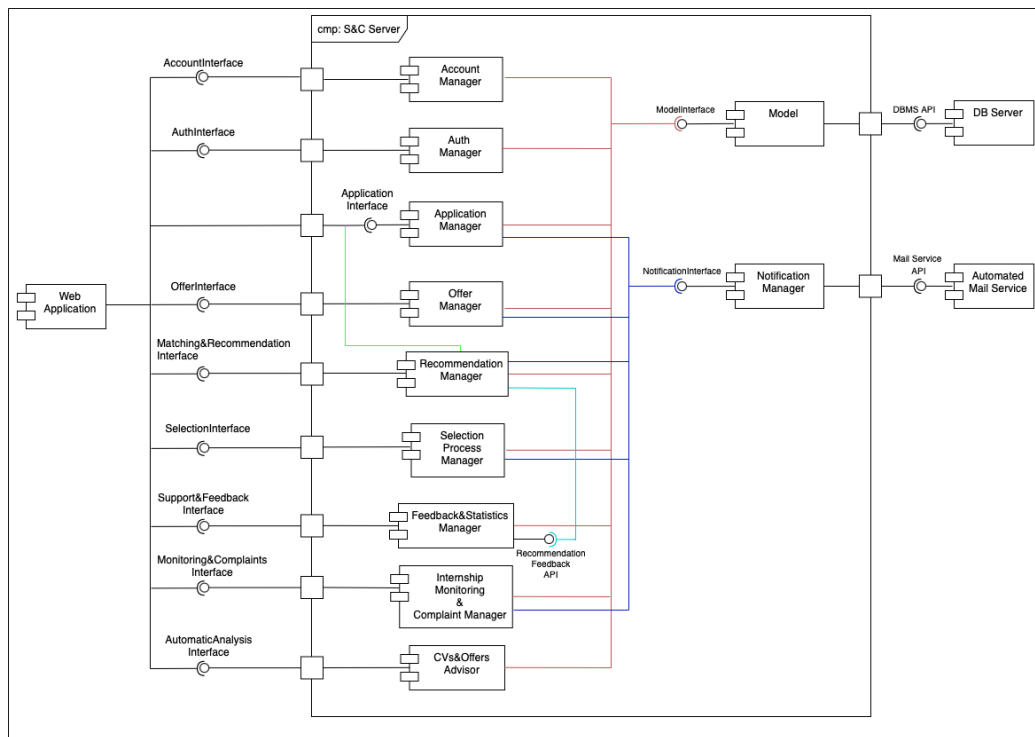
2. Server Side:

- **Web Server:** This server will handle communication with clients over the internet.
- **S&C Server:** This server acts as the intermediary between the Web Server and the database. It can be considered the primary server as it performs data processing tasks, such as the matching algorithm and other complex operations.
- **Database:** Responsible for data storage and management.

2.2 Component View

2.2.1 Students&Companies Server (S&C Server)

Below is the general diagram of the server with an explanation of the external services/servers used through their APIs. The components contained within the server will be explained afterward.



DB server: it contains all the system data, and provides the **DBMS API** interface to allow the S&C server to write, read and delete them.

Automated Mail Service: represents the external server for email automation. It provides the functionality for managing automated emails through the **Mail Service API**.

2.2.2 Model

Is the component that interfaces with the DB Server, so almost every other component in the S&C Server needs to interface with it through the **ModelInterface**, that allows to retrieve, modify and delete data.

2.2.3 Notification Manager

It manages all notifications, both internal to the system and external via email, by interacting with the **Automated Mail Service**, through the external **Mail Service API**. Furthermore, it provides an internal system interface called the **Notification Interface**, which allows other modules to utilise both internal and external notifications. The entire internal messaging system is managed by this module.

2.2.4 Account Manager

This module manages all activities related to each user's account. It handles tasks such as updating personal information, preferences, and other important data necessary for the proper functioning of each individual account, which can be of the following types: Student, Company, or University. Additionally, it provides an interface called **AccountInterface**, which allows users to make changes and manage their S&C account through the web application.

2.2.5 AuthManager

This component manages the login process for all user types. It implements a two-step authentication process:

- **Authentication:** the system prompts the user to enter their username and password to confirm their identity.
- **Authorization:** upon verifying the credentials, the system assigns appropriate permissions to the user based on their role.

It also handles the registration phase.

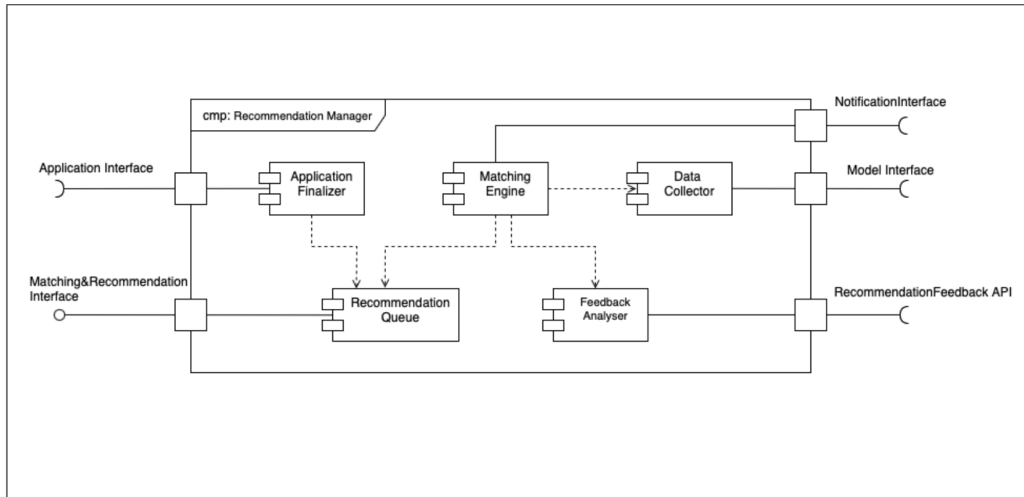
2.2.6 Application Manager

This component is responsible for managing student applications. It handles all aspects of spontaneous applications, withdrawal of an application by a student, and acceptance of an application received through a recommendation (it is important to note that a recommendation officially becomes an application, and is thus recognised as an official application by the company, only when the student decides to accept it). It provides students with the ability to apply for an internship they are interested in through the **Application Interface** on the webApp. It also provides a list of all available internship offers.

2.2.7 Offer Manager

This component is responsible for managing internship postings from companies. It handles all aspects of publishing a internship offer by a company, ensuring that all required fields and details Internship title, internship description, expected duration, internship location, type of internship (curricular/extracurricular), application deadlines are completed without errors or inconsistencies.

2.2.8 Recommendation Manager



This component is responsible for the system's automated recommendation process. It manages the automated evaluation of internship offers provided by companies. After completing the analysis, it suggests internship offers to students based on their preferences, interests, and skills. Additionally, it provides companies with a list of students identified in the preceding phase. It is important to note that, as defined in the concept of automated recommendation, an application via recommendation becomes official for the company only after the student accepts it. Both companies and students can view all information related to the recommendation process on the web app through the **Matching & Recommendation Interface**. The **Recommendation Manager** consists of several internal modules described below:

1. DataCollector

- Collects data extracted from the database, passing through the model.

- Handles data processing and cleaning to ensure the highest accuracy for subsequent analysis.

2. MatchingEngine

- Performs the actual creation of recommendations through a matching process, calculating compatibility using AI/Machine Learning between student profiles and company offers.
- Once the analysis is complete, it selects and generates the actual recommendations.

3. RecommendationQueue

- Stores all recommendations before they become official applications.
- Tracks the status of each recommendation, which can be: *'Pending'*, *'Accepted by Student'*, *'Rejected'* and *'Expired'*.
- Differentiates recommendations that do not convert into applications from those that are officially accepted.

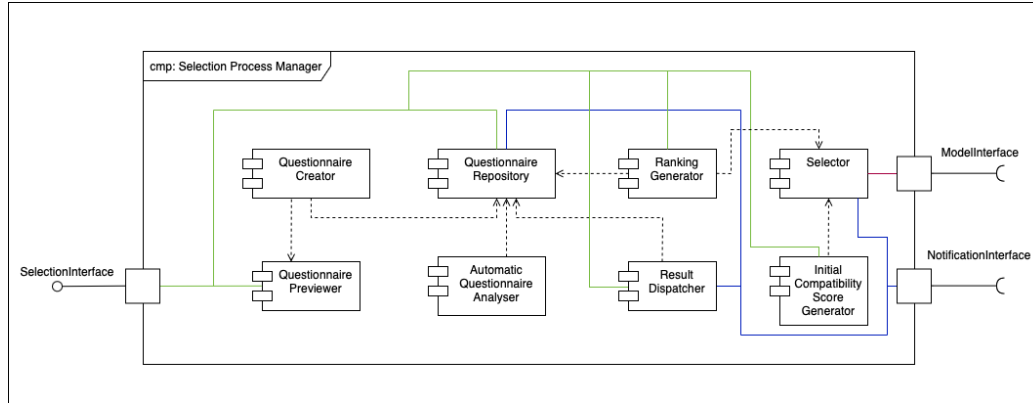
4. Application Finalizer

- Converts a recommendation into an official application only when the student accepts it, updating its status accordingly.
- Interfaces with the **Application Manager** to create an entry for a formal application through the *'Application Interface'*.

5. Feedback Analyser

- Analyses received feedback and improves the matching system. It interfaces with the **Feedback & Statistics Manager** through the **Recommendation Feedback API**.

2.2.9 Selection Process Manager



The **Selection Process Manager** comprises several internal components that work together to enable the entire flow of preparation, approval, compilation, and evaluation of selection questionnaires. It provides users with the above-mentioned functionalities through the '*Selection Interface*'.

- The **Questionnaire Creator** is responsible for generating the questions and structure of each questionnaire, leveraging artificial intelligence and machine learning techniques. It creates questionnaires based on the specific requirements of the company and the types of internships offered.
- The **Questionnaire Previewer** provides a preview of the questionnaire to the company manager, who can assess its relevance and approve it for publication.
- The **Automatic Questionnaire Analyser** evaluates and processes the candidates' responses automatically, producing partial and overall scores for each candidate.
- The **Ranking Generator** helps the company examiner create a ranking of students to facilitate the selection process. It pulls data from the **Questionnaire Repository** and incorporates scores from open-ended responses graded by the examiner, generating an initial ranking that can be confirmed or modified by the examiner. From it, it is possible to accept or reject applications based on the established ranking.
- The **Result Dispatcher** compiles the results into an easily readable format, including open-ended responses that require manual evaluation

by the examiner and any additional information needed by the company to make an informed decision.

- **Questionnaire Repository** manages the storage, retrieval, and updating of questionnaires and their responses, ensuring data is preserved throughout the process. Once the selection process is complete, detailed data is deleted to free up storage space.
- **Selector**: this component is responsible for acquiring all candidates selected by the examiner and saving the activation of the internship in the database. It also saves the non-selected candidates by updating their statuses in the database. Additionally it handles notifying both selected and non-selected candidates.
- **Initial Compatibility Score Generator**: this component is designed to calculate the initial compatibility score provided by the system to the examiner. It helps in filtering out candidates who are not compatible with the internship offer. It thus produces a list containing the following components: the student's first and last name, affiliated university, course of study, the student's CV, and the compatibility score. From it, candidates can be rejected or advanced in the selection process.

2.2.10 Internship Monitoring & Complaint Manager

This component is responsible for managing internships and addressing potential issues or complaints from students or companies. It provides universities with well-formatted data regarding the status of their students' internships and enables students or companies to raise complaints, directly updating the respective statuses of the related internships. All these functionalities are accessible through the **Monitoring & Complaint Interface**.

2.2.11 Feedback & Statistics Manager

This component is responsible for managing user feedback regarding the platform's functionality and the recommendations provided. It offers two interfaces:

- **Support & Feedback Interface**: Allows users to submit complaints or report issues through the webApp.
- **Recommendation Feedback API**: Enables the **Recommendation Manager** module to collect user feedback and suggestions to improve the recommendation process.

2.2.12 CV & Offer Advisor

This component is responsible for providing tailored advice to students and companies to improve their CVs and internship offers, respectively. Using AI and Machine Learning mechanisms, it delivers targeted suggestions to enhance the matching process through recommendations by comparing students' CVs against available company offers. Additionally, it provides companies with specific advice to make their offers as appealing as possible. All these functionalities are made available to users through the **Automatic Analysis Interface**.

2.3 Deployment View

As mentioned earlier, we have decided to opt for a **3-tier** architecture. Each individual layer within the **S&C** platform is managed and structured with dedicated hardware, allowing the system to remain independent from potential issues that could affect any of the layers, as well as enhancing the overall security of the system. Furthermore, this approach enables the platform to be more flexible with regard to potential changes that could be made to parts of the system.

- **Presentation Layer**

This section is dedicated to the application's interaction with the user. Therefore, we will consider the platform's frontend and its interaction with the external world. To communicate with other layers, the platform will use **API** queries over the **HTTPS** protocol, ensuring that the system remains secure, encrypted, and protected against attacks such as data interception (Man-in-the-Middle), thus enhancing the confidentiality and integrity of the exchanged data. All actors will interact with S&C through a web application, which can be accessed via a search engine. Access will be available across different devices (smartphones, tablets, computers) and operating systems (Android, iOS). Devices will interact with the platform through a browser as the client. Therefore, *server-client* and *client-server* requests will be processed and handled using the HTTPS protocol.

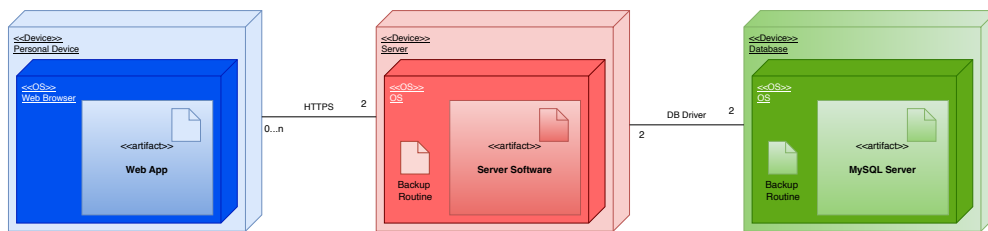
- **Business Logic Layer**

This component contains essential information for the proper functioning of the entire system, as well as the core functionalities of the web application, enabling it to manage business logic and data processing. It handles all internal operations required to fulfill requests from the

Presentation Layer, ensuring that data is processed and returned correctly.

- **Data Management Layer**

This layer contains critical information that allows the entire platform to operate correctly within the environment. Information related to the various actors is managed through a **MySQL** database, with access to the layer facilitated by **API** queries. To prevent potential issues that may arise within the platform or the layer, we have implemented a Runtime-Backup routine to mitigate potential server or database failures. We chose this implementation approach because, for example, if we had opted for server duplication and periodic backups, we could have faced the risk of inconsistencies between the backup copy and the primary database (imagine the significant disruption caused by the loss of some student internship application data, compromising not only the management of registrations but also the entire selection process), or the added complexity of managing synchronization in the strategic planning of the database. Therefore, we have implemented continuous database and server duplication with geographic replication. In this approach, the system performs continuous backups of the data, updating the backup database copy in real-time. The geographically distributed replication of data ensures that copies of the data are stored in different data centers, thus increasing resilience. This diversification provides a level of protection against local disasters, offering high availability and ensuring that the system remains fully operational and accessible to all actors.

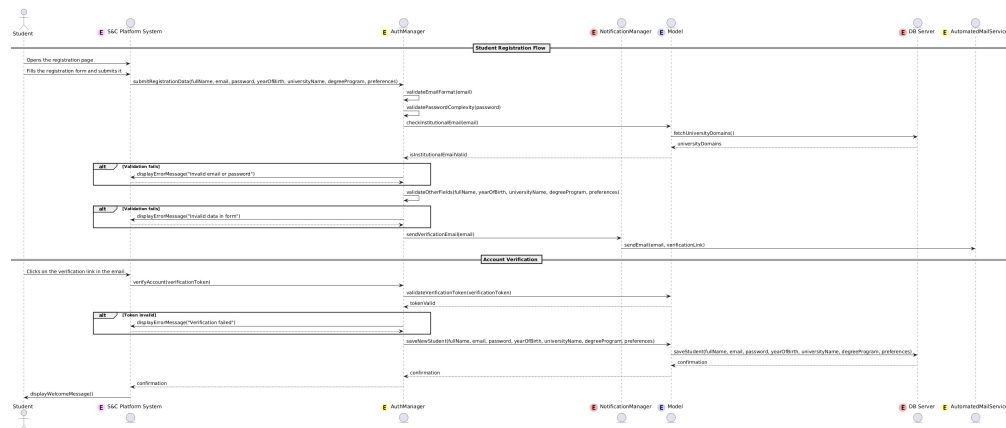


2.4 Runtime View

This section analyzes the interactions between the different components of the system during the execution of the main operations. Each operation is represented through a sequence diagram, which describes the flow of communication between the various modules.

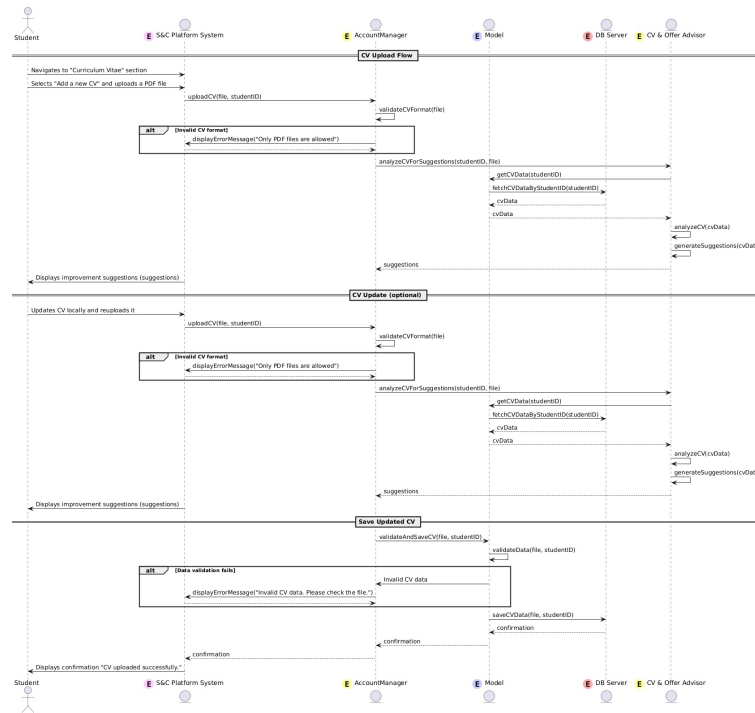
2.4.1 THE STUDENT REGISTERS ON THE PLATFORM

Students can register on the S&C platform by accessing it directly through the **Web Page**. Registration is completed by opening the dedicated page, following the guided enrollment procedure, and entering the required data.



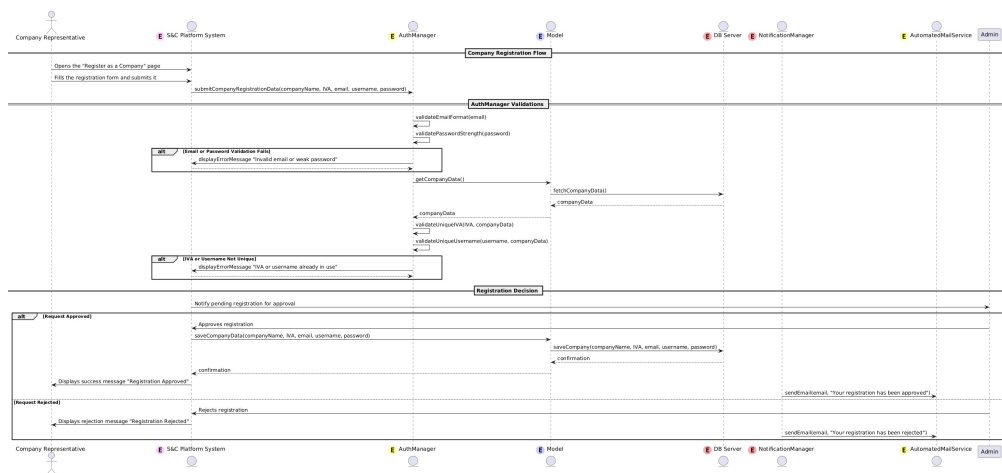
2.4.2 STUDENT'S CV CREATION AND IMPROVEMENT

The **Student** accesses their personal area and, navigating through the portal, reaches the '*Curriculum Vitae*' section, where they can proceed with uploading and editing their CV. During the upload process, suggestions are provided for potential improvements the student can make to make the CV more appealing and targeted to companies.



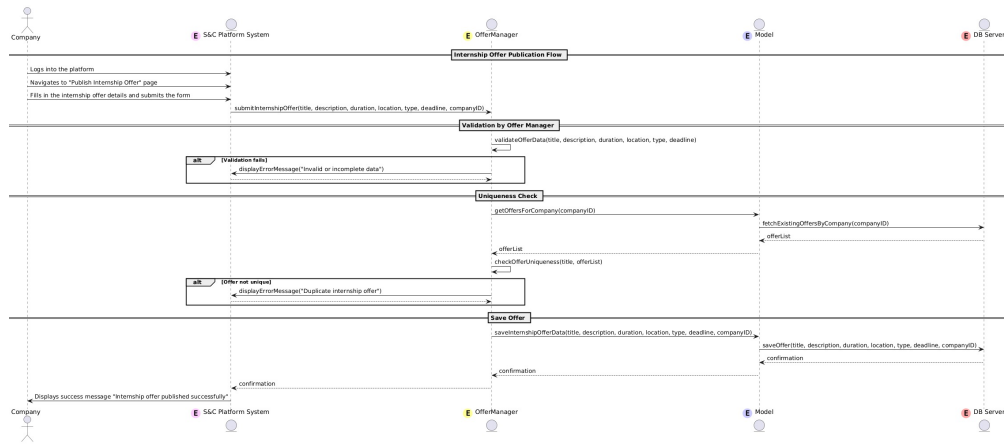
2.4.3 COMPANY REGISTRATION

Company can register on the S&C platform by accessing the dedicated registration section through the Web Page. The registration process involves filling out the required details on the dedicated page and awaiting a notification confirming successful *'Registration Approved'* or *'Registration Rejected'*.



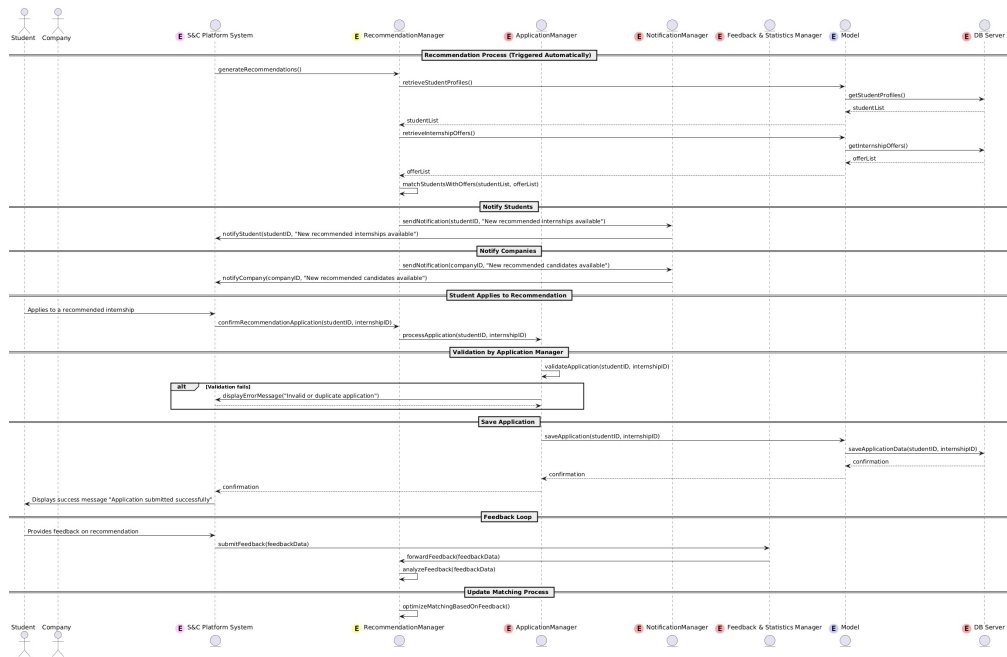
2.4.4 POSTING INTERNSHIP POSITIONS

The **Company** accesses the S&C platform and navigates to the dedicated section for posting a new internship offer: *'Publish Internship Offer'*. This process involves interacting with the database server to store the new offer on the platform, making it available for students to view and apply. The data entered during the creation of the new offer will be validated to ensure accuracy, completeness, and compliance with the required standards.



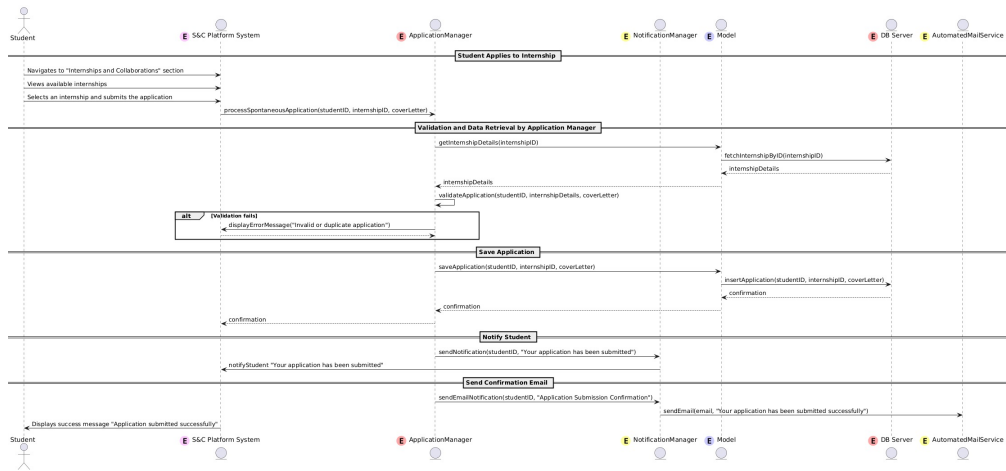
2.4.5 RECOMMENDATION AND MATCHING SYSTEM (MATCHING COMPANIES AND STUDENTS)

The S&C system manages the recommendations and matching process between students and companies. The platform sends notifications to students regarding new recommendations with the message '*New recommended internship available*'. Similarly, companies receive notifications highlighting the availability of new recommended candidates with the message '*New recommended candidates available*'.



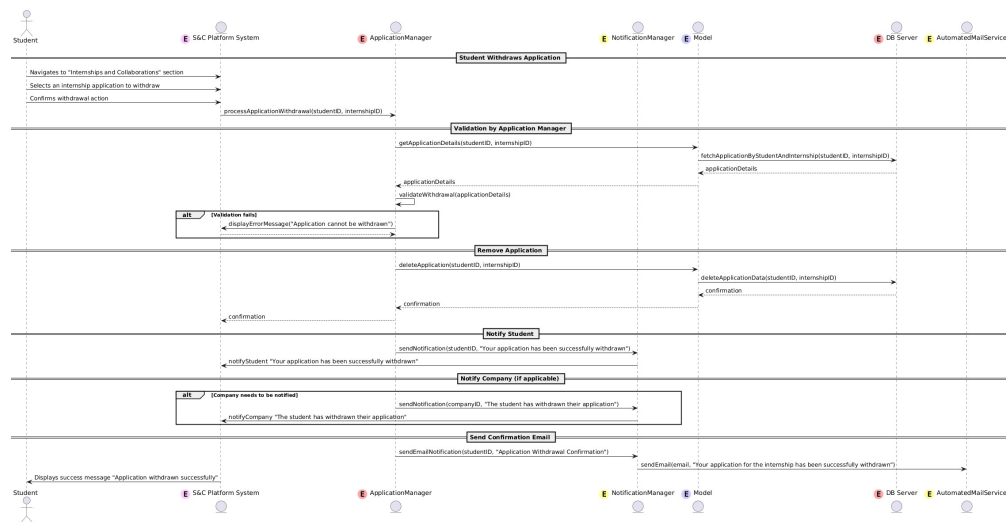
2.4.6 STUDENT'S SPONTANEOUS APPLICATION

By accessing their personal area and selecting '*Internships and Collaborations*', the **Student** can view the available internship opportunities and apply for positions they find interesting. Upon completing the application process, the system will send a notification confirming the successful submission of the application ('*Your application has been submitted*'), followed by an email confirmation ('*Application Submission Confirmation*').



2.4.7 INTERNSHIP WITHDRAWAL AND APPLICATION CANCELLATION PROCESS

By accessing the *'Internships and Collaborations'* section, the **Student** can view the internship offers they have applied for, along with the details of each application. Additionally, the student has the option to withdraw their application from a specific position and will receive a notification and email upon doing so. A notification will be sent to the company associated with the offer only once the application submission period has ended.



2.4.8 INTERNSHIP APPLICATION REVIEW AND SELECTION PROCESS

The **Company**, by accessing the '*Received Applications*' section on the portal, can retrieve all information from the database to view and evaluate the applications received for a specific internship offer. For each candidate, the company has the option to update the application status by selecting either '*Shortlisted*' or '*Rejected*', based on its evaluation criteria. An email notification will be sent to the student as soon as their application is reviewed. If selected, the student will be invited to complete a questionnaire as the next step in the process. After the screening phase, a ranking of the students will be created, determining whether they are selected to participate in the internship with the company.



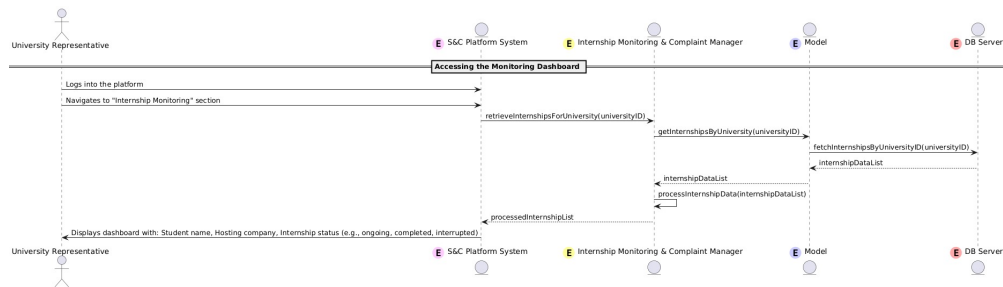
2.4.9 THE STUDENT HAS BEEN SELECTED FOR AN INTERNSHIP IN A COMPANY

The **Student** receives a notification and email regarding the positive outcome of their application for an internship offer, confirming their eligibility to participate in the internship. Through the '*Offer Management*' section, the student can view the status and details of their application. Subsequently, they will have the option to either accept or decline the internship offer. A notification and email will then be sent to the company to communicate the student's decision.



2.4.10 MONITORING OF INTERNSHIPS BY THE UNIVERSITY

The **University** accesses the platform, navigates to the *Internship Monitoring* section, and views the data (stored in **the DB Server**) related to student internships.

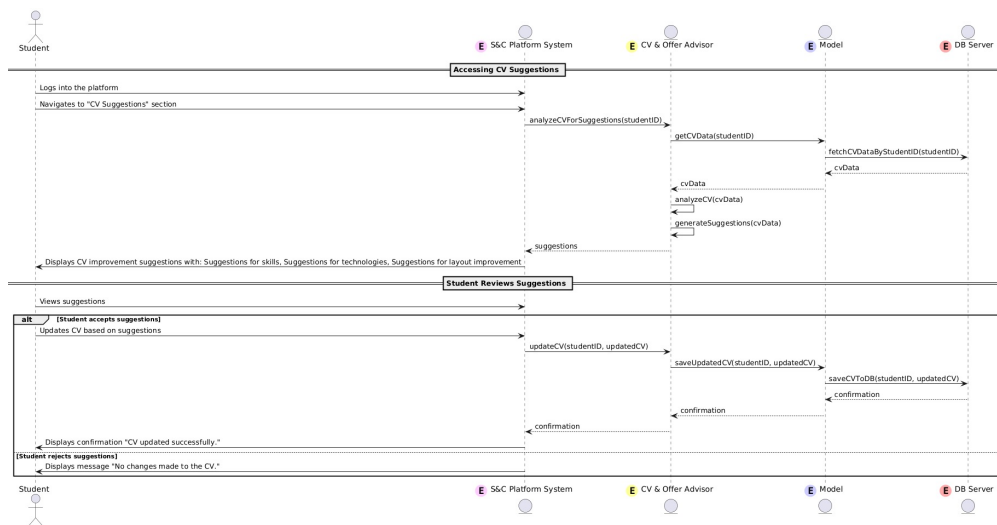


2.4.11 MANAGEMENT OF REPORTS AND COMPLAINTS BY THE UNIVERSITY

To monitor internships, the **University** can access the platform, navigate to the '*Complaint Management*' section, and review specific requests. Additionally, the status of the report will be available. To mediate between the student and the company, the university can contact both parties to resolve any issues. Notifications will be sent to the relevant stakeholders with whom the university wishes to clarify the situation.

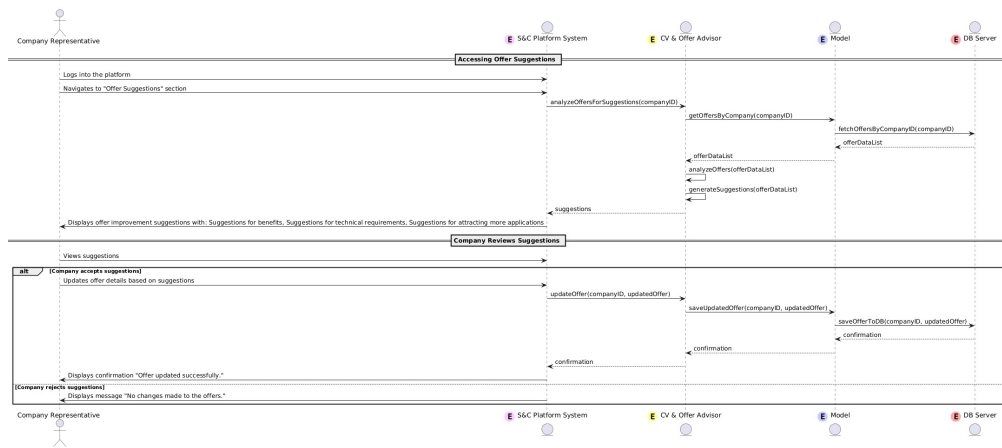
2.4.12 CV IMPROVEMENTS FOR STUDENTS

Student, within their section, will be able to access '*CV Suggestions*' to receive advice on improving their CV. The effectiveness of the student's CV is evaluated through the **CV & Offer Advisor** module. The student may choose to follow the suggestions provided by the platform and update their CV based on the guidelines marked by the system.



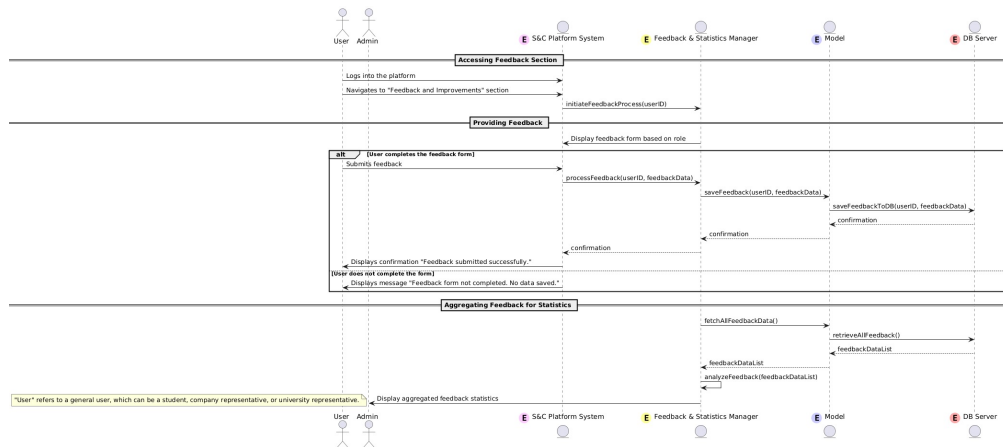
2.4.13 SUGGESTIONS FOR COMPANY

The **Company**, through the dedicated '*Offer Suggestions*' section within the platform, can receive advice via the **CV & Offer Advisor** module, with the aim of making internship offers more attractive, thereby increasing visibility and the number of applications. The company may choose whether or not to follow the suggestions provided by the platform, updating their offers accordingly.



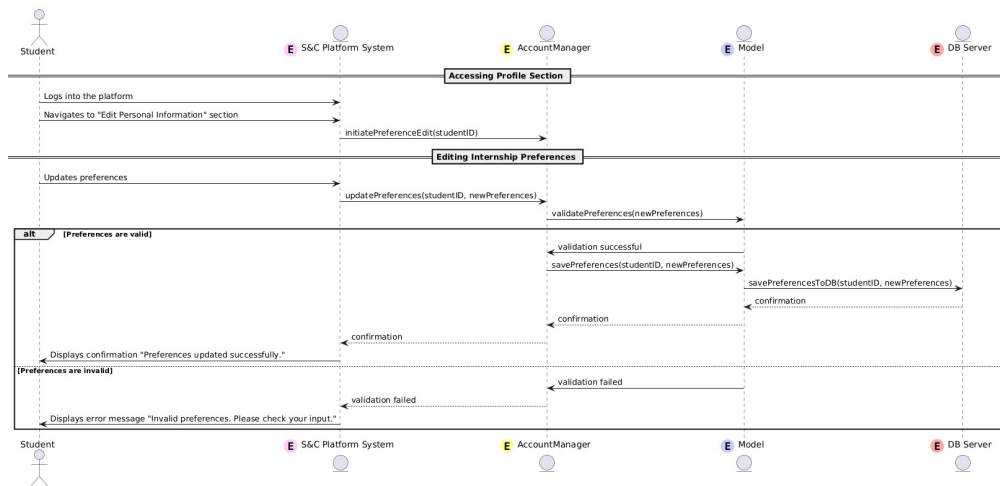
2.4.14 FEEDBACK AND STATISTICS COLLECTION

Users (Students, Companies, Universities) access the S&C platform and select the '*Feedback and Improvements*' option from the main menu. The system presents a customized form to collect feedback based on the user's role. Once completed, the user confirms the submission. The system collects the data and processes it through the **Feedback & Statistic Manager** to generate aggregated statistics, using them to optimize the recommendation system and improve the user experience. Finally, the results are integrated into the learning database to refine future recommendation and matching processes.



2.4.15 MODIFY STUDENT PROFILE TO CHANGE INTERNSHIP PREFERENCES

When the **Student** accesses the platform, within the '*Edit Personal Information*' section, they can modify their preferences, which will be updated in the **DB Server** if valid with the message: '*Preferences updated successfully*'. Otherwise, the message will be: '*Invalid preferences. Please check your input*'.



2.5 Component Interfaces

Details the interfaces between different components in the system.

ModelInterface

- checkInstitutionalEmail(email)
- saveNewStudent(fullName, email, password, yearOfBirth, university-Name, degreeProgram, preferences)
- validateVerificationToken(verificationToken)
- getCVData(studentID)
- validateAndSaveCV(file, studentID)
- validateData(file, studentID)
- getCompanyData()
- getOffersForCompany(companyID)
- saveInternshipOfferData(title, description, duration, location, type, deadline, companyID)
- retrieveStudentProfiles()
- retrieveInternshipOffers()
- saveApplication(studentID, internshipID)
- getInternshipDetails(internshipID)
- saveApplication(studentID, internshipID, coverLetter)
- getApplicationDetails(studentID, internshipID)
- deleteApplication(studentID, internshipID)
- getCandidateData(offerID)
- saveFinalSelection(selectedCandidatesIDList, offerID)
- getInternshipOffer(studentID, offerID)
- updateInternshipStatus(studentID, offerID, newState)

- deleteInternshipApplication(studentID, offerID)
- getInternshipByUniversity(universityID)
- getComplaintForUniversity(universityID)
- getComplaintDetail(supportRequestID)
- saveComplaintStatus(complaintID, status)
- getOfferByCompany(companyID)
- saveUpdatedCV(companyID, updatedCV)
- saveUpdatedOffer(companyID, updatedOffer)
- saveFeedback(userID, feedbackData)
- fetchAllFeedbackData()
- validatePreferences(newPreferences)
- savePreferences(studentID, newPreferences)

DBInterface

- fetchUniversityDomain()
- saveStudent(fullName, email, password, yearOfBirth, universityName, degreeProgram, preferences)
- fetchCVDataByStudentID(studentID)
- saveCVData(file, studentID)
- fetchCompanyData()
- saveCompanyData(companyName, IVA, email, username, password)
- saveCompany(companyName, IVA, email, username, password)
- fetchOffersByCompany(companyID)
- saveOffer(title, description, duration, location, type, deadline, companyID)
- getStudentProfiles()

- getInternshipOffers()
- saveApplicationData(studentID, internshipID)
- fetchInternshipByID(internshipID)
- insertApplication(studentID, internshipID, coverLetter)
- fetchApplicationByStudentAndInternship(studentID, internshipID)
- deleteApplicationData(studentID, internshipID)
- fetchCandidateDataByOfferID(offerID)
- saveSelectionData(selectedCandidatesIDList, offerID)
- fetchInternshipOfferByStudentIDAndOfferID(studentID, offerID)
- updateInternshipStatusInDB(studentID, offerID, newState)
- deleteInternshipApplicationFromDB(studentID, offerID)
- fetchInternshipByUniversity(universityID)
- fetchComplaintDetailsByID(supportRequestID)
- fetchComplaintForUniversity(universityID)
- updateComplaintStatusInDB(complaintID, status)
- saveOfferToDB(companyID, updatedOffer)
- saveFeedbackToDB(userID, feedbackData)
- retrieveFeedbackData()
- savePreferencesToDB(studentID, newPreferences)

AutomatedMailService

- sendEmail(email, content)

NotificationManager

- sendVerificationEmail(email)
- sendNotification(userID, message)
- sendEmailNotification(userID)

AuthManager

- submitRegistrationData(fullName, email, password, yearOfBirth, universityName, degreeProgram, preferences)
- validateEmailFormat(email)
- validatePasswordComplexity(password)
- validateOtherFields()
- verifyAccount(verificationToken)
- submitCompanyRegistrationData(companyName, IVA, email, username, password)
- validateUniqueIVA(IVA, companyData)
- validateUniqueUsername(username, companyData)

AccountManager

- uploadCV(file, studentID)
- validateCVFormat(file)
- initiatePreferenceEdit(studentID)
- updatePreference(studentID, newPreference)

ApplicationManager

- processApplication(studentID, internshipID)
- validateApplication(studentID, internshipID)
- processSpontaneousApplication(studentID, internshipID, coverLetter)
- validateApplication(studentID, internshipDetails, coverLetter)
- processApplicationWithdrawal(studentID, internshipID)
- validateWithdrawal(applicationDetails)

OfferManager

- submitInternshipOffer(title, description, duration, location, type, deadline, companyID)

- validateOfferData(title, description, duration, location, type, deadline)
- checkOfferUniqueness(title, offerList)
- updateCV(studentID, updatedCV)

RecommendationManager

- matchStudentsWithOffers(studentList, offerList)
- confirmRecommendationApplication(studentID, internshipID)
- analyzeFeedback(feedbackData)
- optimizeBasedOnFeedback()
- generateRecommendations()

SelectionProcessManager

- retrieveApplicationsWithInitialScores(offerID)
- calculateInitialScores(candidateData)
- updateCandidateStatus(candidateID, newStatus)
- updateInternalCandidateStatus(candidateID, newStatus)
- createQuestionnairesForShortlisted(offerID, candidateIDs)
- confirmQuestionnaires(offerID)
- markQuestionnairesAsReady(offerID)
- submitQuestionnaireResponse(studentID, offerID, responses)
- analyzeResponsesAndUpdateScores(studentID, responses)
- generateFinalRanking(offerID)
- updateFinalRankings(rankingList)
- updateInternalRanking(rankingList)
- confirmOriginalRanking()
- finalizeRanking()
- finalizeSelectionAndNotify(selectedCandidatesIDList, offerID)

- retrieveInternshipOffer(studentID, offerID)
- acceptInternship(studentID, offerID)
- declineInternship(studentID, offerID)

OfferAdvisorManager

- analyzeOffersForSuggestions(companyID)
- analyzeOffers(offerDataList)
- generateSuggestions(offerDataList)
- updateOffer(companyID, updatedOffer)

InternshipMonitoringComplManager

- retrieveInternshipsForUniversity(universityID)
- processInternshipData(internshipDataList)
- retrieveComplaintForUniversity(universityID)
- retrieveComplaintDetails(complaintID)
- updateComplaintStatus(complaintID, status)

FeedbackStatsManager

- submitFeedback(feedbackData)
- initiateFeedbackProcess(userID)
- processFeedback(userID, feedbackData)
- analyzeFeedback(feedbackData)

CVAdvisorManager

- analyzeCVForSuggestions(studentID, file)
- analyzeCV(cvData)
- generateSuggestions(cvData)

WebApplication

- displayErrorMessage()

- verifyAccount(verificationToken)
- displayWelcomeMessage()
- notifyStudent(studentID, message)
- notifyCompany(companyID, message)
- displayConfirmation(message)
- displayContent(content)

2.6 Selected Architectural Styles and Patterns

Model-View-Controller (MVC):

As deduced from the platform's diagram in Section 2.1, the structure can be divided into three main groups, each fulfilling a distinct macro function. This natural alignment makes the implementation of the **MVC architecture** a logical and effective choice for achieving well-organized and modular software. The **MVC pattern** is a software design paradigm that separates program logic into three distinct components, each assigned a specific responsibility.

- **Model:** The model is independent of the user interface and serves as the central component that manages data from the database. This part operates on the server side.
- **View:** The view is the part that allows users to interact with other components. Through the various pages of the S&C platform, users can interact with the model by inputting and examining data.
- **Controller:** The controller is responsible for performing all operations related to data processing. On the platform, this is where components such as the **Account Manager**, **Auth Manager**, **Application Manager**, **Offer Manager**, **Recommendation Manager**, **Selection Process Manager**, **Feedback & Statistics Manager**, **Internship Monitoring & Complaint Manager** and **CV & Offers Advisor** (described in Chapter 2.1) will reside.

Three-Tier Architecture:

As described in Section 2.3, the S&C application uses a three-tier architecture that separates the presentation, server, and database layers into three distinct hardware components. Since the S&C application is a distributed system, this architecture is chosen to better accommodate different types of clients. The server and database layers will reside on the server, while the

presentation layer will run on client devices. The client will be a thick client, as the presentation layer will include data validation to offload heavy workloads—such as data storage and complex processing—to the server.

Client-Server Architecture:

In this architecture, the client and server are logically separated and act as distinct entities. They communicate over a network, where the client sends requests, and the server responds. This architecture is used to manage connections and facilitate the exchange of information between each client and the database.

2.7 Other Design Decisions

In this section, other decisions related to the project are presented and explained.

2.7.1 Availability

To ensure the handling of a high volume of requests while maintaining the required performance, one or more load balancers can be introduced to manage incoming request distribution. This approach allows requests to be managed effectively and optimizes performance.

2.7.2 Security

To ensure and enhance the security of the system, firewalls are introduced at the following points:

- Between web portals and the web server
- Between the web server and the S&C server
- Between the S&C server and the database
- Between the S&C server and the automated email system

3 User Interface Design

This section will present the user interface of the S&C platform, developed as a web application. The presentation will focus on the features analyzed and described in the previous chapters. The appearance and characteristics of the graphical interface will vary based on the user's profile, adapting to the specific type of user.



Figure 1: FirstPage - User

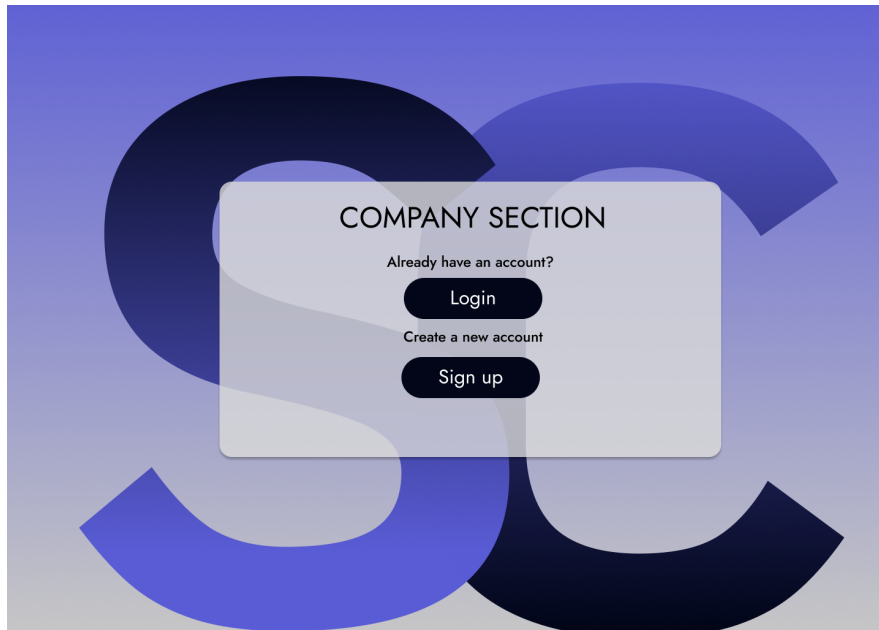


Figure 2: FirstPage - Company

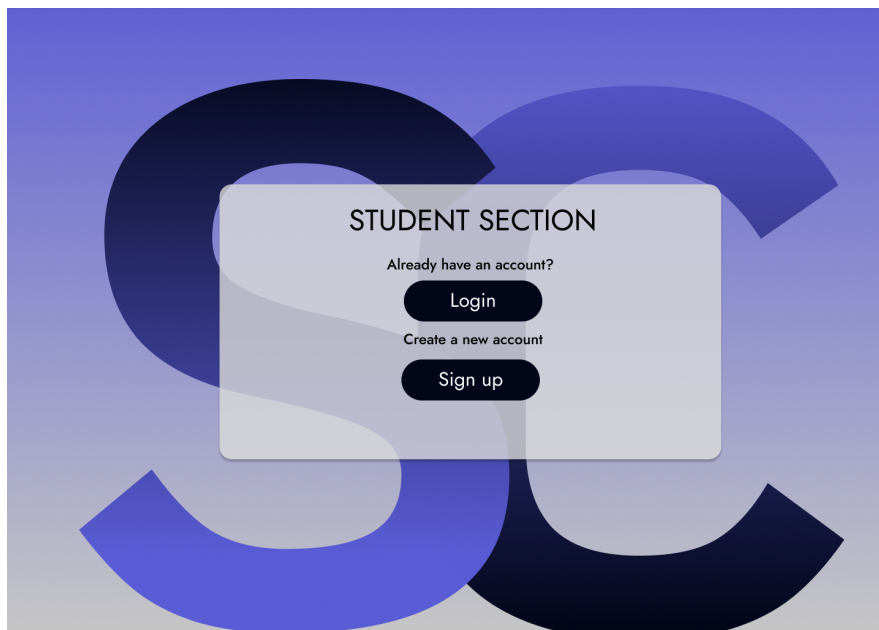
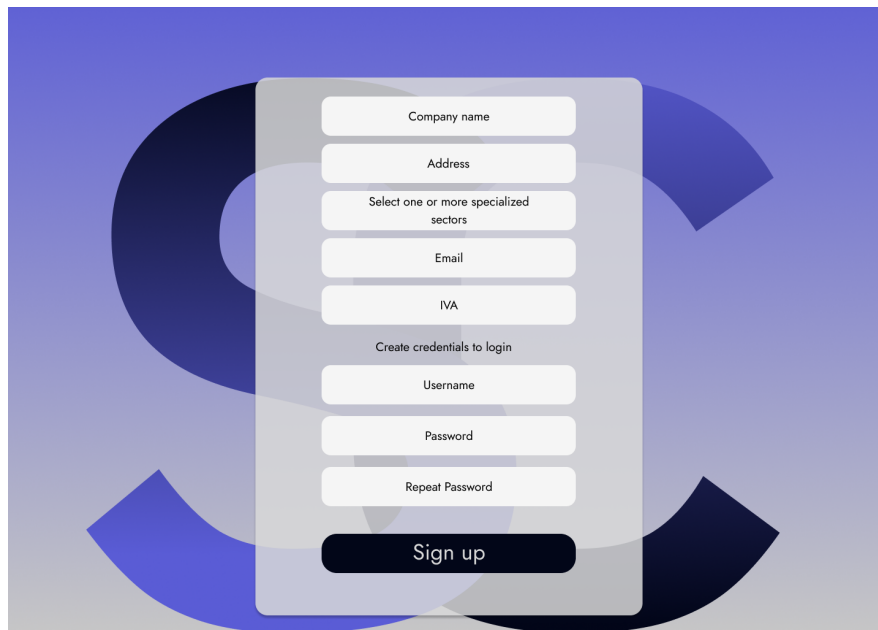


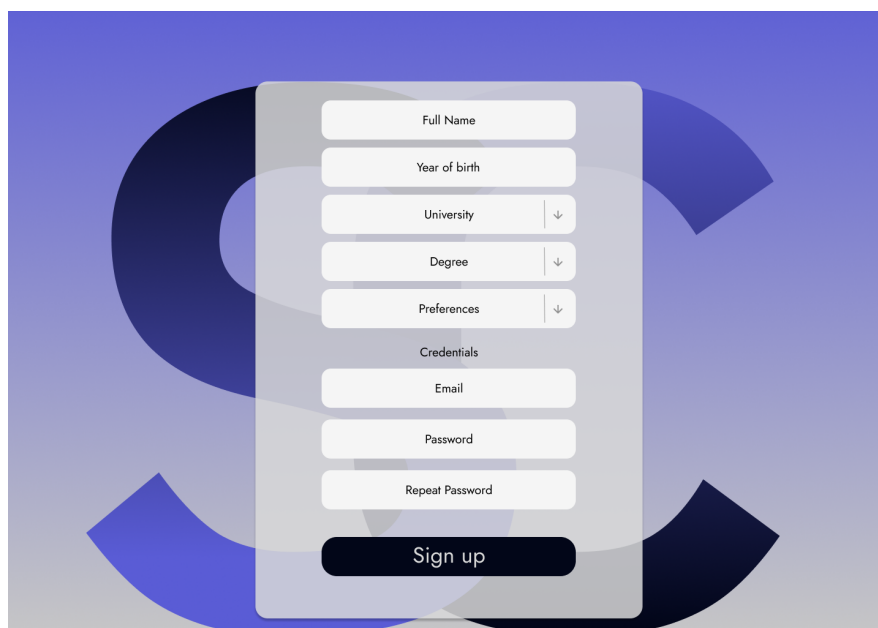
Figure 3: FirstPage - Student



A sign-up form for companies. The form is centered on a blue background with a large, stylized 'C' and two curved arrows. The form fields are as follows:

- Company name
- Address
- Select one or more specialized sectors
- Email
- IVA
- Create credentials to login
- Username
- Password
- Repeat Password
- Sign up

Figure 4: Sign Up - Company



A sign-up form for students. The form is centered on a blue background with a large, stylized 'C' and two curved arrows. The form fields are as follows:

- Full Name
- Year of birth
- University
- Degree
- Preferences
- Credentials
- Email
- Password
- Repeat Password
- Sign up

Figure 5: Sign Up - Student

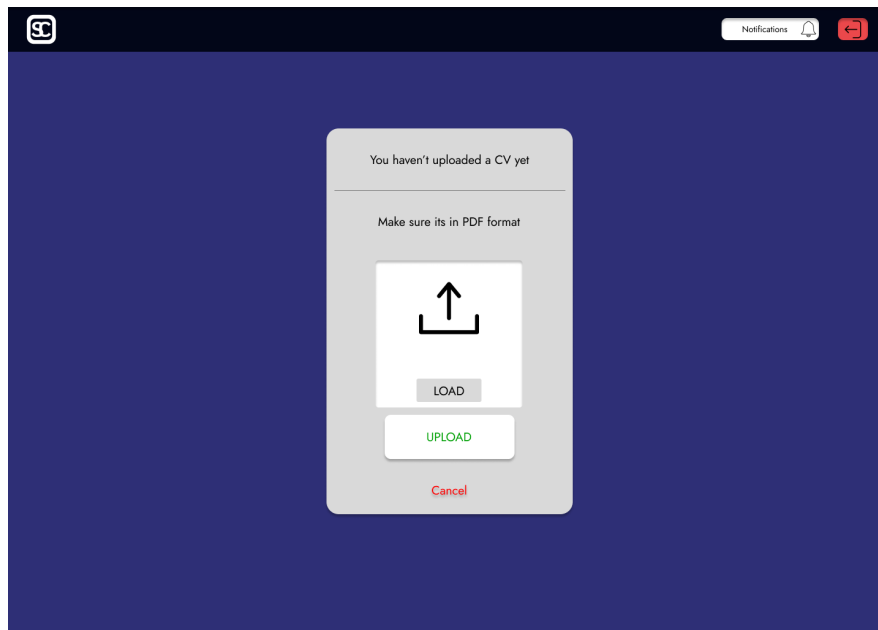


Figure 6: Curriculum Vitae Section – Adding Phases for a New CV - Student

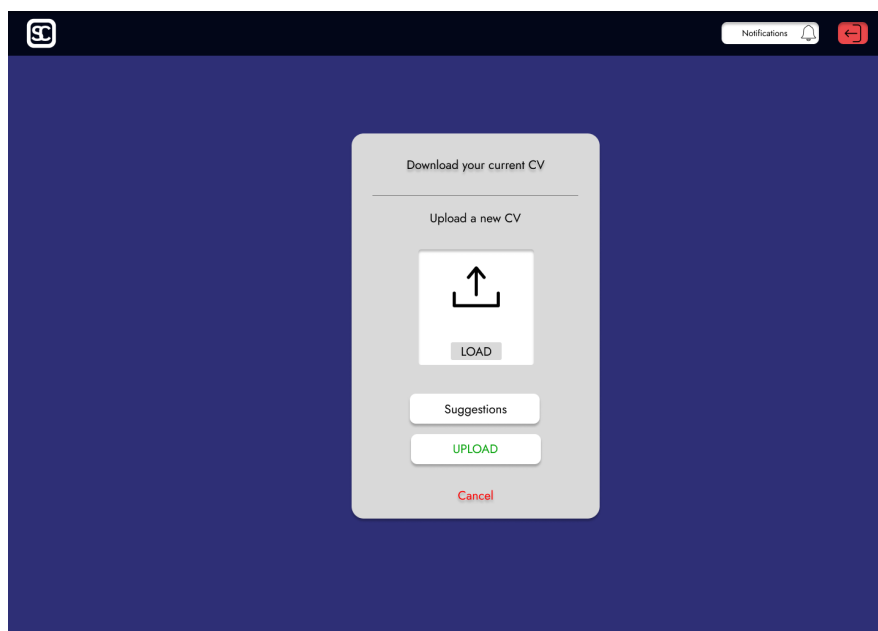


Figure 7: Curriculum Vitae Section - Edit - Student

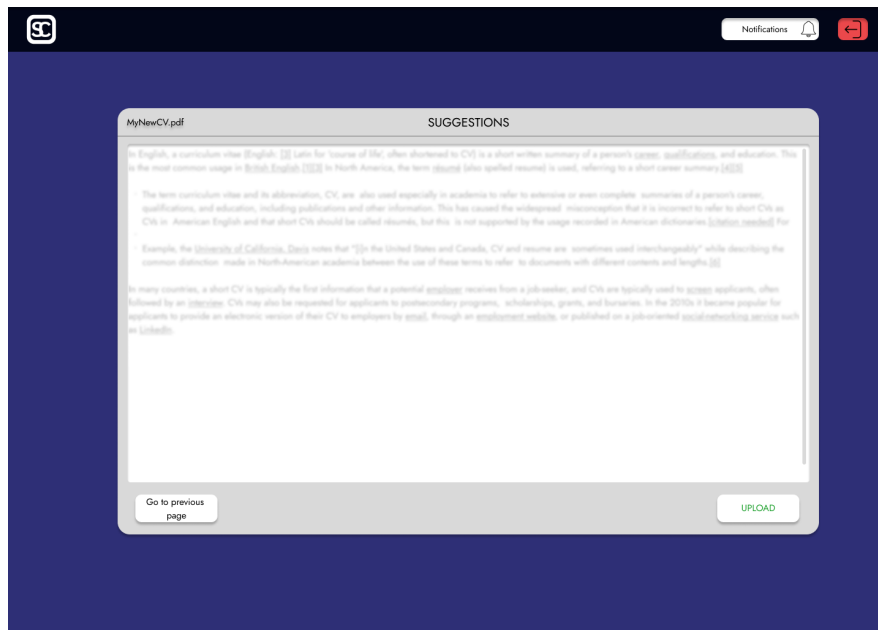


Figure 8: CV Suggestions - Student

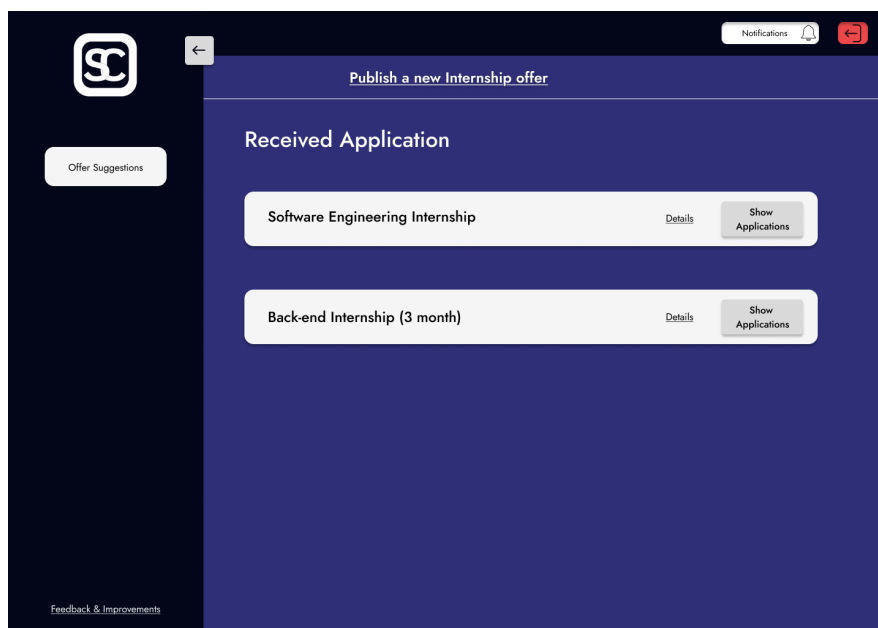


Figure 9: Received Application - Company

The screenshot shows a modal form titled "Offer Details" on a dark blue background. The form has a light gray background and contains the following fields and buttons:

- Title:** A single-line text input field.
- Description:** A multi-line text area.
- Duration:** A single-line text input field.
- Location:** A single-line text input field.
- Type:** A single-line text input field.
- Deadline:** A single-line text input field with a dropdown arrow.
- Buttons:** A white "Cancel" button and a dark blue "Publish Internship Offer" button.

Figure 10: Publish Internship Offer (details) - Company

The screenshot shows the "Internships and Collaborations" page for a student. The page has a dark blue background and features the following elements:

- Header:** A dark blue header with a logo, a "Notifications" button, and a red button.
- Section Header:** "Internships and Collaborations" in white text.
- Available Internships:** A list of four internship cards, each with a company logo, name, "Details" link, and "Apply" link.
 - Connection Inc.
 - Technology SPA
 - CarFactory
 - Company, Inc.
- Notifications:** A dropdown menu showing two notifications:
 - "New recommended internship at Company, Inc. available" with "Details" and "Apply" links.
 - "New recommended internship at CarFactory available" with "Details" and "Apply" links.

Figure 11: Internships and Collaborations - Student

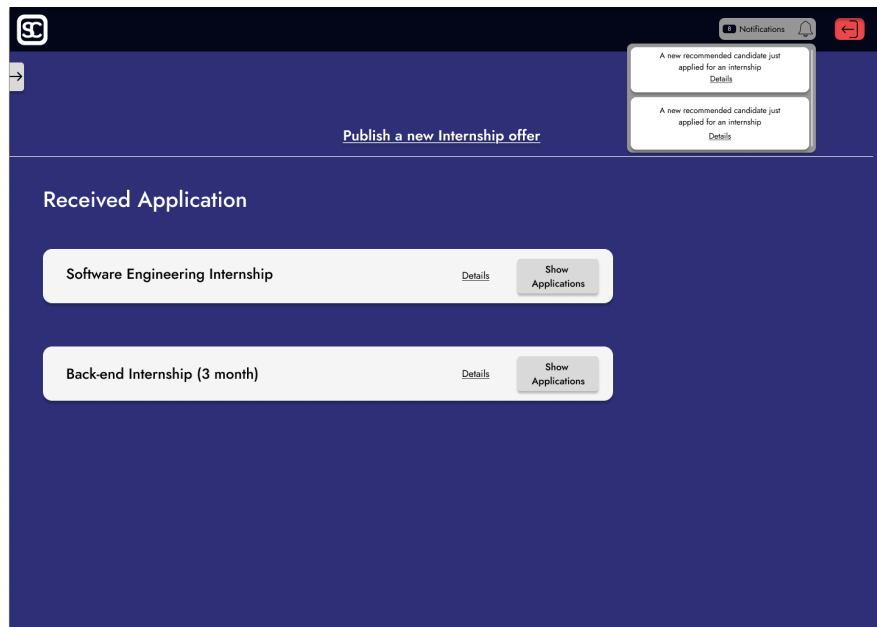


Figure 12: Received Application (notification) - Company

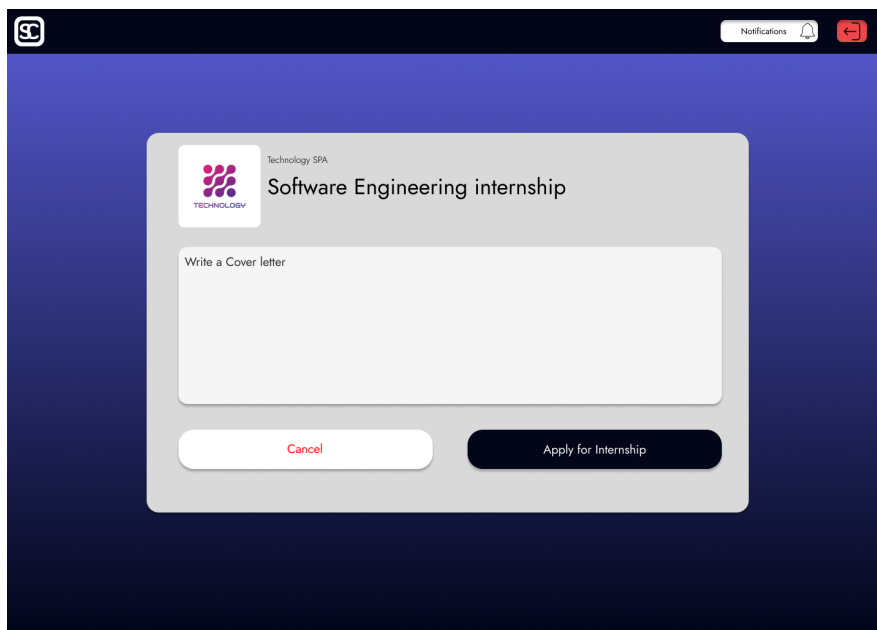


Figure 13: Spontaneous Application (cover letter) - Student

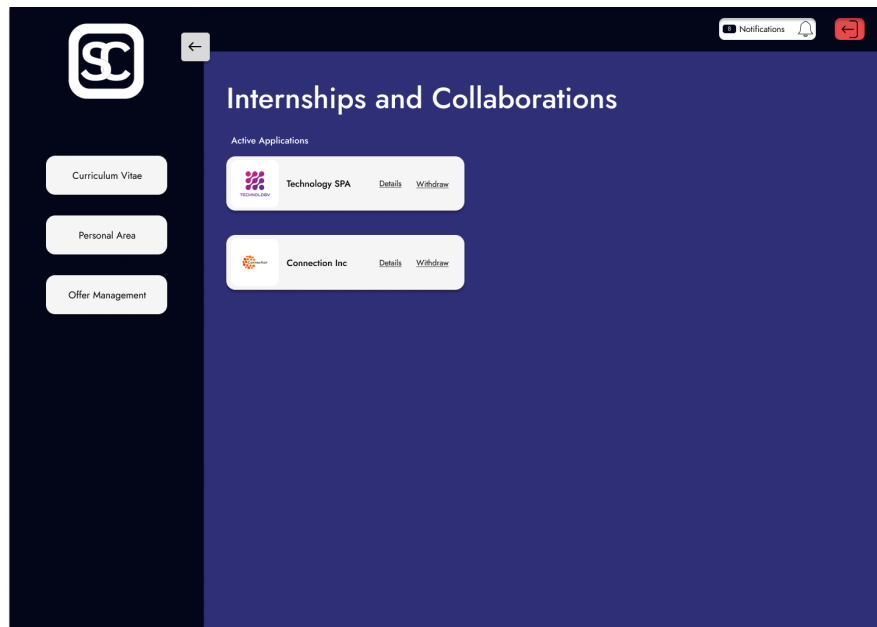


Figure 14: MainPageStud - Student

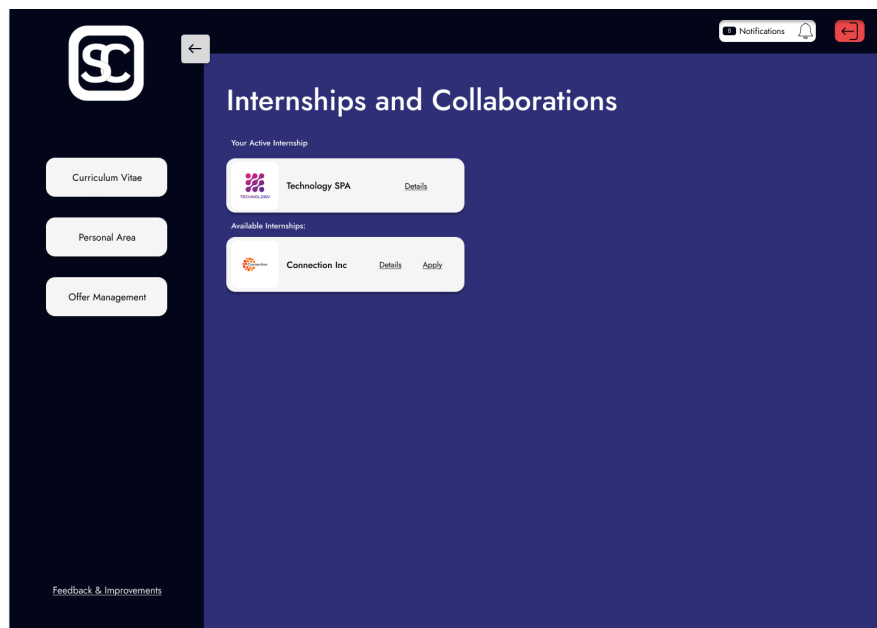


Figure 15: Recommendation and Matching system - Student

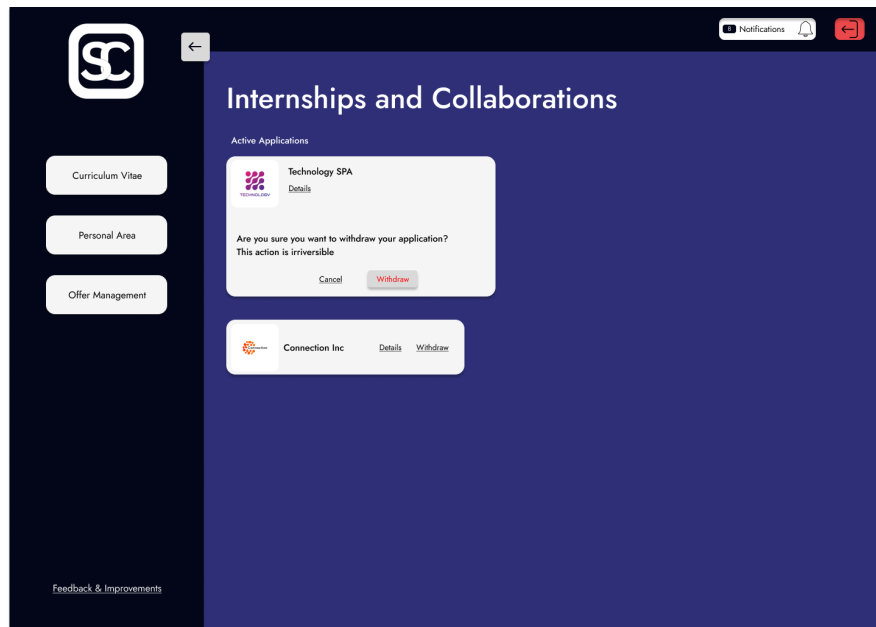


Figure 16: Internship Withdrawal and Application Cancellation Process - Student

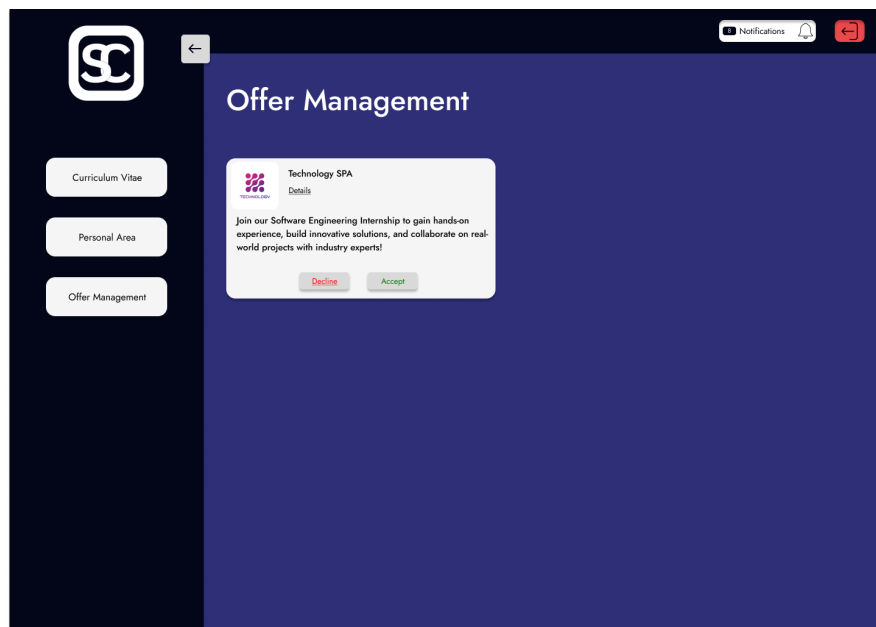


Figure 17: Offer Management - Student

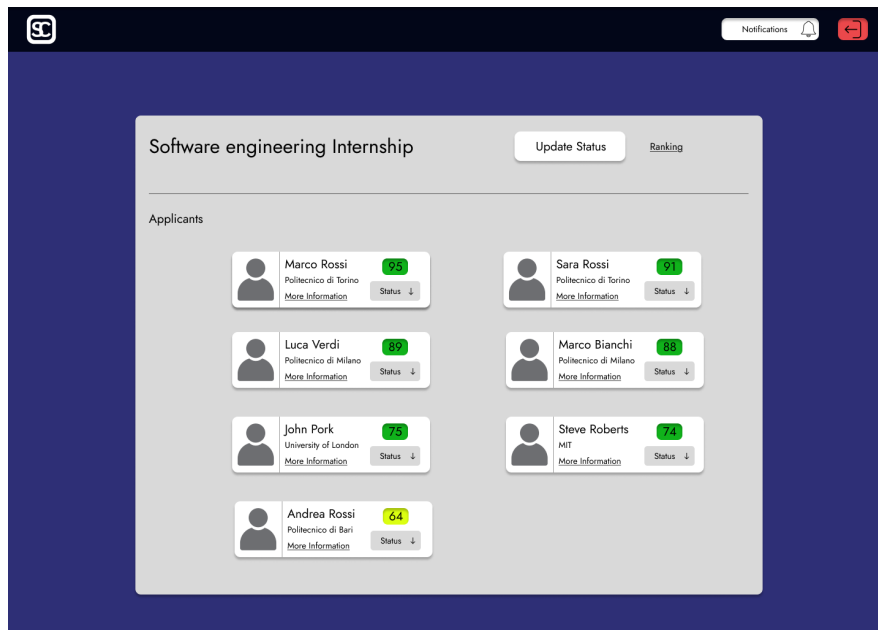


Figure 18: Report related to an Ongoing Internship - University

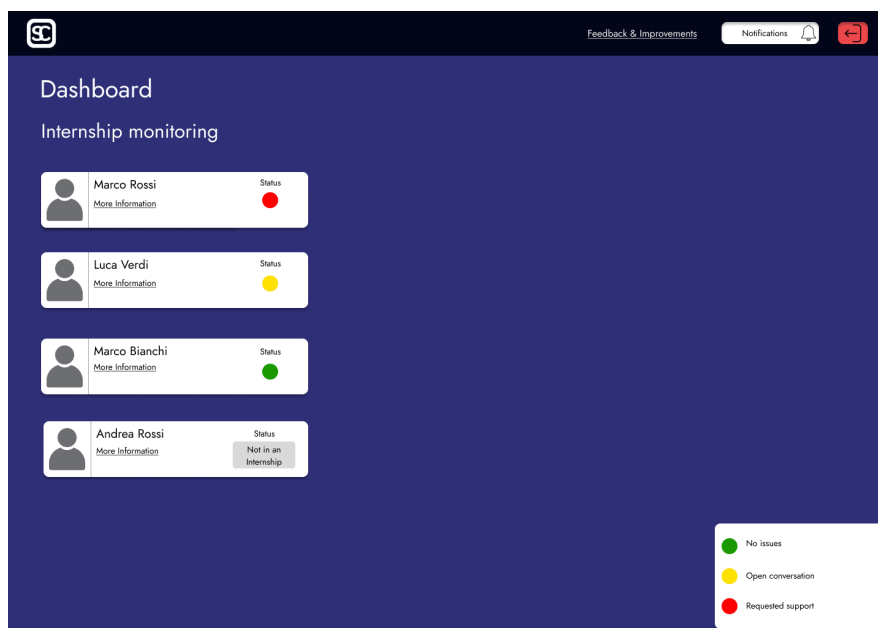


Figure 19: Internship Monitoring - University

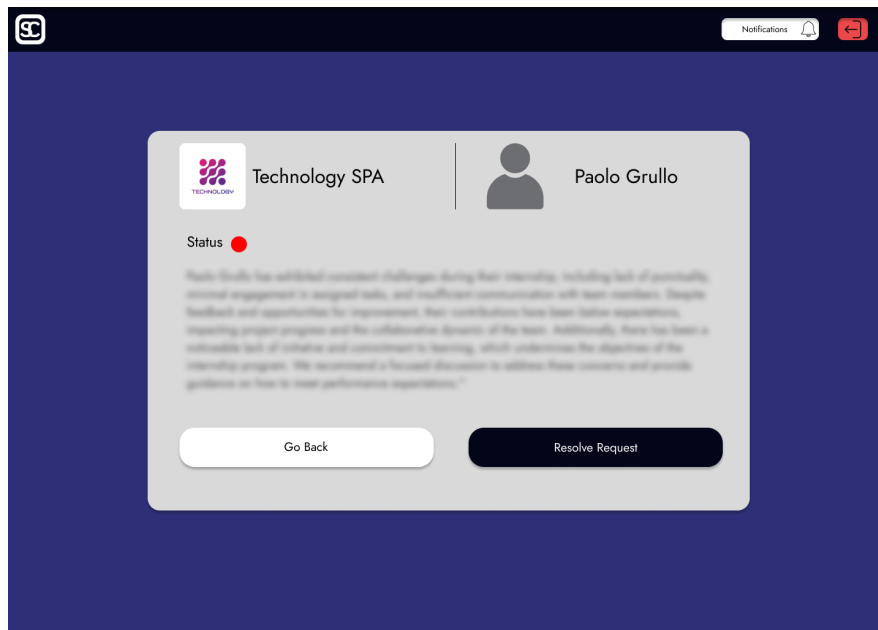


Figure 20: Internship Monitoring (red.status)- University

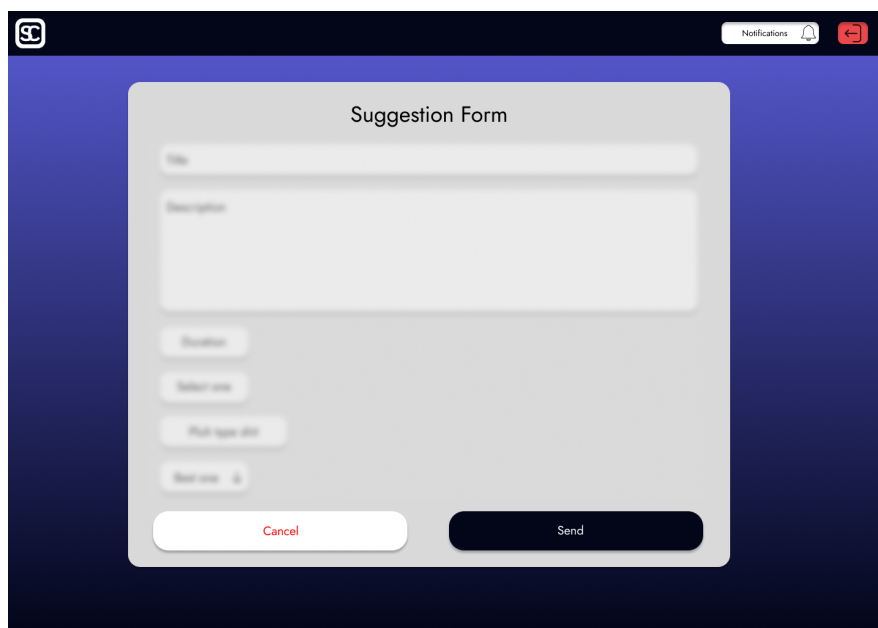


Figure 21: Feedback and Improvements - All User

4 Requirements Traceability

This section provides an overview of how the requirements defined in the RASD are systematically mapped to the design elements detailed in this document. The mapping ensures that each requirement is addressed during the design phase, maintaining consistency and traceability between the analysis and design stages.

To facilitate the understanding and tracking of the requirements identified in the RASD with respect to the functions designed in the DD, the Requirements are presented below.

Table 2: Mapping of Requirements to Components

Req	M	NM	AM	AutM	AppM	OM	RM	SPM	IC	FaS	CVOA
R1	✓			✓							
R2	✓			✓							
R3	✓			✓							
R4	✓			✓							
R5	✓			✓							
R6	✓			✓							
R7	✓			✓							
R8	✓		✓								
R9	✓		✓								
R10	✓		✓								
R11	✓										✓
R12	✓					✓					
R13	✓										✓
R14	✓		✓								
R15	✓								✓		
R16	✓						✓				
R17	✓	✓					✓				
R18	✓	✓					✓				
R19							✓			✓	
R20	✓				✓						
R21	✓				✓						
R22	✓					✓					
R23								✓			
R24								✓			
R25		✓						✓			
R26	✓				✓		✓				
R27	✓							✓			
R28	✓								✓		
R29	✓								✓		
R30	✓	✓							✓		
R31	✓	✓							✓		
R32	✓								✓		
R33	✓								✓		
R34	✓								✓		
R35		✓									
R36		✓									
R37		✓									

1.Registration and Identification

<i>ID</i>	<i>Description</i>
R1	The system must allow students to register using truthful, detailed personal information.
R2	The system must allow companies to register on the platform using contact information and details about the company's mission.
R3	The system must allow universities to register on the platform after entering into an agreement with S&C.
R4	The system must enable students to verify their identity to access the system through the sign-in operation.
R5	The system must enable companies to verify their identity to access the system through the sign-in operation.
R6	The system must enable universities to verify their identity to access the system through the sign-in operation.
R7	The system must restrict access to the platform's functionalities for unauthenticated users.

2.User Profile Management

<i>ID</i>	<i>Description</i>
R8	The system must allow students to update their personal profile.
R9	The system must allow students to upload their CV in PDF format.
R10	The system must allow students to view and update their CV, including displaying the date it was last updated.
R11	The system must provide suggestions for improving the students' CVs.
R12	The system must allow companies to create, modify, and publish internship offers.
R13	The system must provide suggestions for improving internship offers to companies.

<i>ID</i>	<i>Description</i>
R14	The system must allow companies to modify the contact information of the internship coordinator.
R15	The system must allow universities to monitor the progress of internships.

3. Matching and Recommendation System

<i>ID</i>	<i>Description</i>
R16	The system must perform matching between student profiles and internship offers.
R17	The system must allow students to receive recommendations for relevant offers.
R18	The system must allow companies to receive suggestions for suitable students for their offers.
R19	The system must allow users and companies to provide feedback on the recommendations.

4. Application and Selection Management

<i>ID</i>	<i>Description</i>
R20	The system must allow students to apply for internship offers through the platform.
R21	The system must allow students to withdraw their application.
R22	The system must allow companies to view the applications received.
R23	The system must allow companies to evaluate candidates through structured questionnaires.
R24	The system must collect responses to the questionnaires and present the results to companies.
R25	The system must allow companies to communicate the outcome of the applications.

<i>ID</i>	<i>Description</i>
R26	The system must allow students to accept or reject the offers received.

5. Internship Monitoring and Management

<i>ID</i>	<i>Description</i>
R27	The system must update the internship status to “Active” after acceptance.
R28	The system must allow universities, students, and companies to monitor the progress of the internship.
R29	The system must allow students and companies to report issues or complaints.
R30	The system must notify the university corresponding to the student when a complaint is present.
R31	The system must allow the university to contact the student and the company to resolve any issues arising during the internship.
R32	The system must allow companies to provide feedback on the students’ performance during and after the internship.
R33	The system must allow students to provide feedback on their internship experience.
R34	The system must manage the early termination of the internship, if requested by the university, through a defined procedure involving all parties.

6.Communication and Notifications

<i>ID</i>	<i>Description</i>
R35	The system must provide an internal messaging feature for notifications and important alerts.
R36	The system must be able to send email notifications for important events.
R37	The system must integrate with external services for automated email sending.

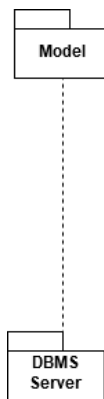
5 Implementation, Integration and Test Plan

5.1 General Implementation

This section provides a general overview of the implementation process, adopting a bottom-up approach to streamline the testing phase and align it more closely with the development phase. The system is divided into several modules, and the integration of these components will be detailed in the following subsection.

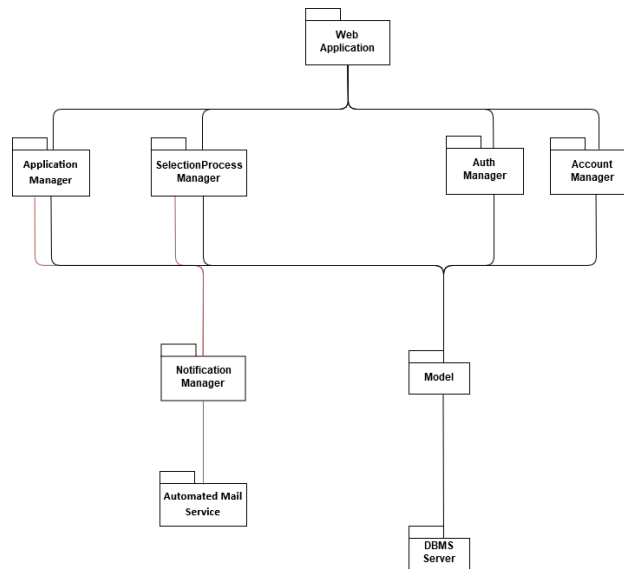
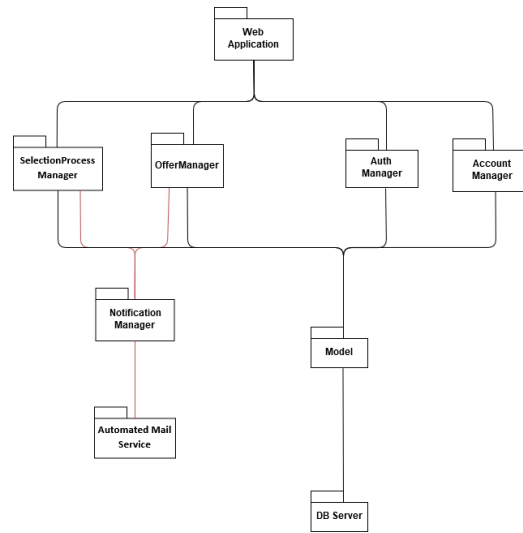
5.2 Database Integration

This is strongly recommended as the first phase, as everything revolves around storing, updating, and deleting information. Therefore, the initial step is to establish the connection between the model and the database.



5.2.1 Authentication

In the second phase, authentication components are implemented to focus on login mechanics. This step is still part of the server-side development, and the client connection, which is not yet present, can be simulated during this phase.



5.2.3 Student and Company Advanced Functionalities

After developing the basic functionalities in the previous step, the focus can shift to more advanced features. Specifically, modules such as the CV & Offers Advisor, Recommendation Manager, and Feedback & Statistics Manager should be developed for both clients. With these additions, the platform will be complete with all the defined functionalities.

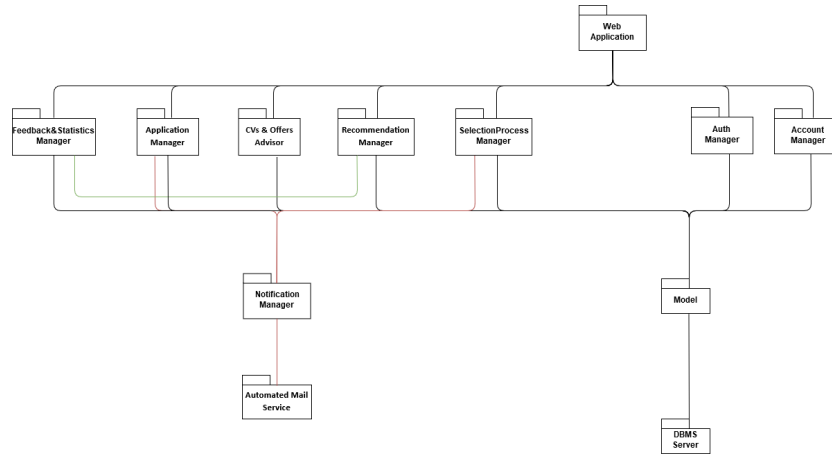


Figure 22: Student

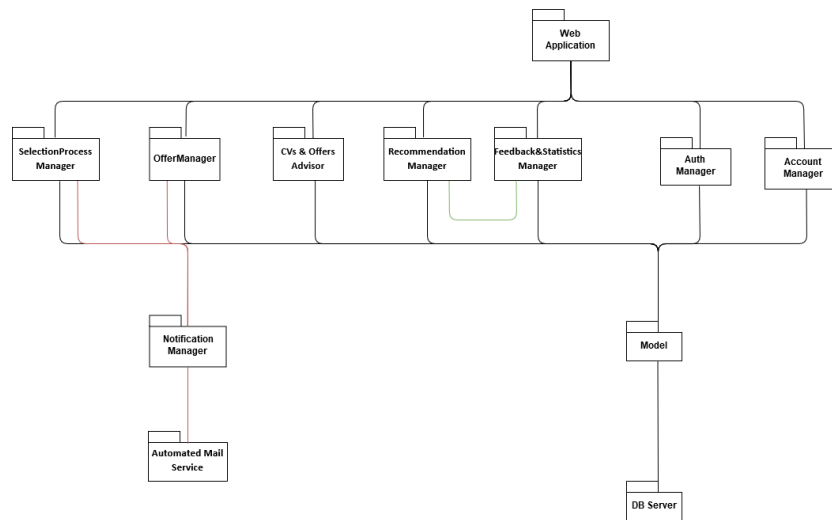


Figure 23: Company

5.3 Test Plan

5.3.1 General Testing

The testing process should proceed in alignment with the program's development, following a *step-by-step* approach. Each function must be tested before moving on to the implementation of the next one. Additionally, it is crucial to test all dependencies to ensure that each component integrates correctly. This implies that testing must cover all connections and interactions represented in the diagrams above.

5.3.2 AI Training

Testing the Recommendation Engine and CV & Offer Advisor

Supervised learning modules will be implemented to train the integrated AI system, providing large datasets used to develop, improve, and train all the components where it is involved. For example, in the automated matching phase or CV's analysis.

6 Effort Spent

Nardi Andrea

Activity	Hours Spent
Introduction	1
Architectural Design	13
User Interface Design	8
Requirements Traceability	3
Implementation, Integration, and Test Plan	4

Pesaturo Christian Giovanni

Activity	Hours Spent
Introduction	2
Architectural Design	11
User Interface Design	8
Requirements Traceability	4
Implementation, Integration, and Test Plan	4

Pinessi Andrea

Activity	Hours Spent
Introduction	1
Architectural Design	10
User Interface Design	12
Requirements Traceability	2
Implementation, Integration, and Test Plan	4